

theoretical of mix-budget

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1 Complete Proofs for the Approximation Analysis

This document provides complete proofs for the approximation analysis of the mixed-budget attention scope construction. We first show that the sparse attention approximation error is controlled by the missing attention mass. We then show how random walks reduce the expected missing mass through hitting probabilities. Finally, we prove the mixed-budget error bound for the candidate set constructed from both real graph edges and random-walk-generated edges.

1.1 Restricted Softmax and Missing Mass

Let $G = (V, E)$ be the input graph and let $q \in V$ be a query node. Full attention allows q to attend to all nodes in V . For each candidate node $v \in V$, let $s_{qv} \in R$ denote the attention score and define

$$w_{qv} = \exp(s_{qv}).$$

The full softmax normalizer and attention distribution are

$$Z_q = \sum_{u \in V} w_{qu}, \quad \alpha_{qv} = \frac{w_{qv}}{Z_q}, \quad v \in V.$$

Given value vectors $V_v \in R^d$, the full attention output is

$$y_q = \sum_{v \in V} \alpha_{qv} V_v.$$

For a sparse candidate set $S_q \subseteq V$, the restricted softmax normalizer and attention distribution are

$$Z_{q,S} = \sum_{u \in S_q} w_{qu}, \quad \tilde{\alpha}_{qv} = \frac{w_{qv}}{Z_{q,S}}, \quad v \in S_q.$$

The sparse attention output is

$$\tilde{y}_q = \sum_{v \in S_q} \tilde{\alpha}_{qv} V_v.$$

[Restricted softmax as conditional distribution] For any $v \in S_q$,

$$\tilde{\alpha}_{qv} = \frac{\alpha_{qv}}{\sum_{u \in S_q} \alpha_{qu}}.$$

Equivalently, $\tilde{\alpha}_{qv}$ is the full attention distribution conditioned on the event that the sampled key lies in S_q .

For any $v \in S_q$, we have

$$\tilde{\alpha}_{qv} = \frac{w_{qv}}{Z_{q,S}}.$$

Multiplying both the numerator and denominator by $1/Z_q$ gives

$$\tilde{\alpha}_{qv} = \frac{w_{qv}/Z_q}{Z_{q,S}/Z_q}.$$

By definition, $w_{qv}/Z_q = \alpha_{qv}$. Moreover,

$$\frac{Z_{q,S}}{Z_q} = \frac{\sum_{u \in S_q} w_{qu}}{Z_q} = \sum_{u \in S_q} \frac{w_{qu}}{Z_q} = \sum_{u \in S_q} \alpha_{qu}.$$

Therefore,

$$\tilde{\alpha}_{qv} = \frac{\alpha_{qv}}{\sum_{u \in S_q} \alpha_{qu}}.$$

This is exactly the conditional distribution of $u \sim \alpha_q$ given $u \in S_q$.

[Error controlled by missing attention mass] Assume $\|V_v\| \leq M$ for all $v \in V$. Define the missing attention mass of S_q as

$$\rho_q(S_q) = \sum_{v \in V \setminus S_q} \alpha_{qv} = 1 - \sum_{v \in S_q} \alpha_{qv}.$$

Then

$$\|\tilde{y}_q - y_q\| \leq 2M\rho_q(S_q).$$

If $\rho_q(S_q) = 0$, then the full attention distribution assigns zero mass to $V \setminus S_q$. In this case, the restricted softmax distribution over S_q is identical to the full attention distribution on its support, and therefore $\tilde{y}_q = y_q$. The claimed bound holds trivially.

Now assume $\rho_q(S_q) > 0$. For brevity, write $\rho_q = \rho_q(S_q)$. Define the conditional mean of value vectors inside S_q as

$$\mu_S = \sum_{v \in S_q} \frac{\alpha_{qv}}{1 - \rho_q} V_v,$$

and define the conditional mean of value vectors outside S_q as

$$\mu_{S^c} = \sum_{v \in V \setminus S_q} \frac{\alpha_{qv}}{\rho_q} V_v.$$

By the restricted-softmax conditional distribution lemma, the sparse attention output is exactly the conditional mean inside S_q :

$$\tilde{y}_q = \sum_{v \in S_q} \tilde{\alpha}_{qv} V_v = \sum_{v \in S_q} \frac{\alpha_{qv}}{\sum_{u \in S_q} \alpha_{qu}} V_v = \sum_{v \in S_q} \frac{\alpha_{qv}}{1 - \rho_q} V_v = \mu_S.$$

The full attention output can be decomposed into the contribution from S_q and the contribution from its complement:

$$y_q = \sum_{v \in S_q} \alpha_{qv} V_v + \sum_{v \in V \setminus S_q} \alpha_{qv} V_v.$$

Using the definitions of μ_S and μ_{S^c} , this becomes

$$y_q = (1 - \rho_q) \mu_S + \rho_q \mu_{S^c}.$$

Therefore,

$$y_q - \tilde{y}_q = (1 - \rho_q) \mu_S + \rho_q \mu_{S^c} - \mu_S = \rho_q (\mu_{S^c} - \mu_S).$$

Taking norms gives

$$\|y_q - \tilde{y}_q\| = \rho_q \|\mu_{S^c} - \mu_S\| \leq \rho_q (\|\mu_{S^c}\| + \|\mu_S\|).$$

Since μ_S and μ_{S^c} are convex combinations of value vectors $\{V_v\}$ and $\|V_v\| \leq M$ for all v , we have

$$\|\mu_S\| \leq M, \quad \|\mu_{S^c}\| \leq M.$$

Thus,

$$\|y_q - \tilde{y}_q\| \leq \rho_q (M + M) = 2M\rho_q.$$

This completes the proof.

1.2 Random-Walk Missing Mass

Let S_q^{rw} be the candidate set generated by random walks from the query node q . Specifically, generate R independent random walks of length L :

$$X_0^{(r)} = q, \quad X_{t+1}^{(r)} \sim P(X_t^{(r)}, \cdot), \quad r = 1, \dots, R, \quad t = 0, \dots, L-1,$$

where P is the transition matrix of G . The random-walk candidate set is

$$S_q^{\text{rw}} = \bigcup_{r=1}^R \{X_t^{(r)} : t = 0, 1, \dots, L\}.$$

For each node $v \in V$, define the length- L hitting probability from q to v as

$$h_q(v; L) = \Pr(\exists t \in \{0, \dots, L\} : X_t = v \mid X_0 = q).$$

[Random-walk missing mass] For the random-walk candidate set S_q^{rw} ,

$$E [\rho_q(S_q^{\text{rw}})] = \sum_{v \in V} \alpha_{qv} (1 - h_q(v; L))^R.$$

Consequently, under the value bound $\|V_v\| \leq M$,

$$E [\|\tilde{y}_q - y_q\|] \leq 2M \sum_{v \in V} \alpha_{qv} (1 - h_q(v; L))^R.$$

Fix a node $v \in V$. For one random walk starting from q , the probability that v is visited within length L is $h_q(v; L)$. Therefore, the probability that this single walk does not visit v is

$$1 - h_q(v; L).$$

Because the R random walks are generated independently, the probability that none of the R walks visits v is

$$\Pr(v \notin S_q^{\text{rw}}) = (1 - h_q(v; L))^R.$$

The missing attention mass of S_q^{rw} is

$$\rho_q(S_q^{\text{rw}}) = \sum_{v \in V \setminus S_q^{\text{rw}}} \alpha_{qv}.$$

Equivalently, using the indicator function,

$$\rho_q(S_q^{\text{rw}}) = \sum_{v \in V} \alpha_{qv} \mathbf{1}\{v \notin S_q^{\text{rw}}\}.$$

Taking expectation and using linearity of expectation, we obtain

$$E [\rho_q(S_q^{\text{rw}})] = \sum_{v \in V} \alpha_{qv} \Pr(v \notin S_q^{\text{rw}}).$$

Substituting the missing probability gives

$$E [\rho_q(S_q^{\text{rw}})] = \sum_{v \in V} \alpha_{qv} (1 - h_q(v; L))^R.$$

Finally, by the missing-mass error lemma,

$$\|\tilde{y}_q - y_q\| \leq 2M \rho_q(S_q^{\text{rw}}).$$

Taking expectation on both sides yields

$$E [\|\tilde{y}_q - y_q\|] \leq 2M E [\rho_q(S_q^{\text{rw}})].$$

Substituting the expression for the expected missing mass completes the proof.

1.3 Mixed-Budget Approximation Bound

We now consider the mixed-budget candidate set, which combines selected real graph edges and random-walk-generated candidates. Let B be the total per-query budget. The budget is divided into a deterministic real-edge budget r and a stochastic random-walk budget w , where

$$r + w \leq B.$$

The mixed candidate set is

$$S_q(r, w) = S_q^{\text{real}}(r) \cup S_q^{\text{rw}}(w).$$

Here $S_q^{\text{real}}(r)$ denotes the candidate nodes selected through real graph edges, and $S_q^{\text{rw}}(w)$ denotes the candidate nodes generated through random walks under budget w . Let $R(w)$ and $L(w)$ denote the number and length of random walks under budget w .

[Mixed-budget attention approximation] Assume $\|V_v\| \leq M$ for all $v \in V$. Let $A_q \subseteq V$ be a high-mass set satisfying

$$\mu_q(A_q) = \sum_{v \in A_q} \alpha_{qv} \geq 1 - \eta_q,$$

where $\eta_q \in [0, 1)$. Define the real-edge coverage over A_q as

$$\gamma_q(r) = \sum_{v \in A_q} \alpha_{qv} \mathbf{1}\{v \in S_q^{\text{real}}(r)\}.$$

Define the minimum hitting probability over the remaining high-mass nodes as

$$h_{\min, q}(w) = \min_{v \in A_q \setminus S_q^{\text{real}}(r)} h_q(v; L(w)).$$

Then the expected approximation error of the mixed-budget sparse attention output satisfies

$$E \left[\|\tilde{y}_q^{(r, w)} - y_q\| \right] \leq 2M [\eta_q + (\mu_q(A_q) - \gamma_q(r)) \exp(-R(w)h_{\min, q}(w))].$$

By the missing-mass error lemma, the sparse attention approximation error is controlled by the missing attention mass. Therefore,

$$E \left[\|\tilde{y}_q^{(r, w)} - y_q\| \right] \leq 2ME [\rho_q(S_q(r, w))].$$

It remains to upper bound the expected missing mass of the mixed candidate set.

For a fixed node $v \in V$, the event $v \notin S_q(r, w)$ occurs if v is selected by neither the real-edge component nor the random-walk component. Since the real-edge component is deterministic after $S_q^{\text{real}}(r)$ is chosen, we have

$$\mathbf{1}\{v \notin S_q^{\text{real}}(r)\} = 1 - \mathbf{1}\{v \in S_q^{\text{real}}(r)\}.$$

For the random-walk component, the probability that v is not hit by any of the $R(w)$ independent walks of length $L(w)$ is

$$(1 - h_q(v; L(w)))^{R(w)}.$$

Therefore,

$$\Pr(v \notin S_q(r, w)) = (1 - \mathbf{1}\{v \in S_q^{\text{real}}(r)\}) (1 - h_q(v; L(w)))^{R(w)}.$$

Using the indicator form of missing mass, we have

$$\rho_q(S_q(r, w)) = \sum_{v \in V} \alpha_{qv} \mathbf{1}\{v \notin S_q(r, w)\}.$$

Taking expectation gives

$$E[\rho_q(S_q(r, w))] = \sum_{v \in V} \alpha_{qv} \Pr(v \notin S_q(r, w)).$$

Substituting the missing probability yields

$$E[\rho_q(S_q(r, w))] = \sum_{v \in V} \alpha_{qv} (1 - \mathbf{1}\{v \in S_q^{\text{real}}(r)\}) (1 - h_q(v; L(w)))^{R(w)}.$$

We split this sum into the high-mass set A_q and its complement $V \setminus A_q$:

$$E[\rho_q(S_q(r, w))] = T_1 + T_2,$$

where

$$T_1 = \sum_{v \in A_q} \alpha_{qv} (1 - \mathbf{1}\{v \in S_q^{\text{real}}(r)\}) (1 - h_q(v; L(w)))^{R(w)}$$

and

$$T_2 = \sum_{v \in V \setminus A_q} \alpha_{qv} (1 - \mathbf{1}\{v \in S_q^{\text{real}}(r)\}) (1 - h_q(v; L(w)))^{R(w)}.$$

We first bound T_2 . Since both multiplicative terms are at most 1, we have

$$T_2 \leq \sum_{v \in V \setminus A_q} \alpha_{qv}.$$

By the definition of A_q ,

$$\sum_{v \in A_q} \alpha_{qv} = \mu_q(A_q) \geq 1 - \eta_q.$$

Therefore,

$$\sum_{v \in V \setminus A_q} \alpha_{qv} \leq \eta_q,$$

and hence

$$T_2 \leq \eta_q.$$

We now bound T_1 . For any node $v \in A_q$ that is already included in $S_q^{\text{real}}(r)$, we have

$$1 - \mathbf{1}\{v \in S_q^{\text{real}}(r)\} = 0.$$

Thus, such nodes do not contribute to T_1 . Therefore, T_1 can be written as

$$T_1 = \sum_{v \in A_q \setminus S_q^{\text{real}}(r)} \alpha_{qv} (1 - h_q(v; L(w)))^{R(w)}.$$

By the definition of $h_{\min,q}(w)$, for every $v \in A_q \setminus S_q^{\text{real}}(r)$,

$$h_q(v; L(w)) \geq h_{\min,q}(w).$$

Therefore,

$$1 - h_q(v; L(w)) \leq 1 - h_{\min,q}(w).$$

Since the function $x \mapsto x^{R(w)}$ is monotone increasing on $[0, 1]$,

$$(1 - h_q(v; L(w)))^{R(w)} \leq (1 - h_{\min,q}(w))^{R(w)}.$$

Using the inequality $1 - x \leq \exp(-x)$ for $x \in [0, 1]$, we further have

$$(1 - h_{\min,q}(w))^{R(w)} \leq \exp(-R(w)h_{\min,q}(w)).$$

Thus,

$$T_1 \leq \exp(-R(w)h_{\min,q}(w)) \sum_{v \in A_q \setminus S_q^{\text{real}}(r)} \alpha_{qv}.$$

It remains to express the remaining mass in A_q after real-edge coverage. By definition,

$$\mu_q(A_q) = \sum_{v \in A_q} \alpha_{qv},$$

and

$$\gamma_q(r) = \sum_{v \in A_q} \alpha_{qv} \mathbf{1}\{v \in S_q^{\text{real}}(r)\} = \sum_{v \in A_q \cap S_q^{\text{real}}(r)} \alpha_{qv}.$$

Therefore,

$$\sum_{v \in A_q \setminus S_q^{\text{real}}(r)} \alpha_{qv} = \mu_q(A_q) - \gamma_q(r).$$

Substituting this into the bound for T_1 gives

$$T_1 \leq (\mu_q(A_q) - \gamma_q(r)) \exp(-R(w)h_{\min,q}(w)).$$

Combining the bounds on T_1 and T_2 , we obtain

$$E[\rho_q(S_q(r, w))] \leq \eta_q + (\mu_q(A_q) - \gamma_q(r)) \exp(-R(w)h_{\min,q}(w)).$$

Finally, applying the missing-mass error lemma gives

$$E[\|\tilde{y}_q^{(r,w)} - y_q\|] \leq 2M[\eta_q + (\mu_q(A_q) - \gamma_q(r)) \exp(-R(w)h_{\min,q}(w))].$$

This completes the proof.